ECON 483 FINAL PROJECT

DOES ACADEMIC PAPER TITLE COMPLEXITY INFLUENCE HOW OFTEN A PAPER IS CITED?

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Abstract

This paper examines whether the stylistic complexity of academic paper titles affects citation outcomes. Using an original dataset of 231 economics papers from OpenAlex and a replicated sample of nearly 800 papers from Semantic Scholar, I analyze title length, average word length, and punctuation features such as colons and question marks. Citation counts are modeled as log(citations + 1) and estimated using Ordinary Least Squares with standard controls. Across both datasets, title length, word complexity, and colon usage show no meaningful relationship with citation counts. However, in the larger dataset, question-style titles are associated with significantly fewer citations. Publication year is the strongest predictor of citations, reflecting differences in exposure time. Overall, the results suggest that title complexity plays a limited role in determining academic impact.

1. Introduction

Selecting an effective title is an important but often underestimated aspect of academic writing. Titles serve as the primary signal of a paper's content and may influence whether potential readers choose to engage with the research. In economics, where scholarly output is large and highly competitive, even small stylistic choices, such as title length or the use of punctuation, could plausibly affect a paper's visibility and citation performance. Despite frequent discussion among researchers about what makes a "good" title, there is limited empirical evidence on whether title complexity meaningfully influences citation outcomes.

This paper examines whether the stylistic complexity of academic paper titles is associated with citation counts in economics. I focus on observable title characteristics, including the number of words in the title, average word length, and the use of colons or question marks. Citation counts are used as a measure of academic impact [1]. The central question is whether these stylistic features are systematically related to citations after accounting for basic differences across papers [5].

To answer this question, I estimate a series of regression models using two independently collected datasets. The analysis is designed to assess whether results are consistent across samples rather than driven by a particular dataset or specification. Across models, the findings suggest that while authors may devote significant attention to crafting titles, most stylistic features have limited influence on academic impact. Instead, citation outcomes appear to be driven more by structural factors than by title complexity alone.

2. Data

This paper uses two datasets of economics papers to study the relationship between title complexity and citation outcomes. The first dataset serves as the main analysis, while the second dataset is used to assess robustness and ensure that results are not driven by sample size or data source.

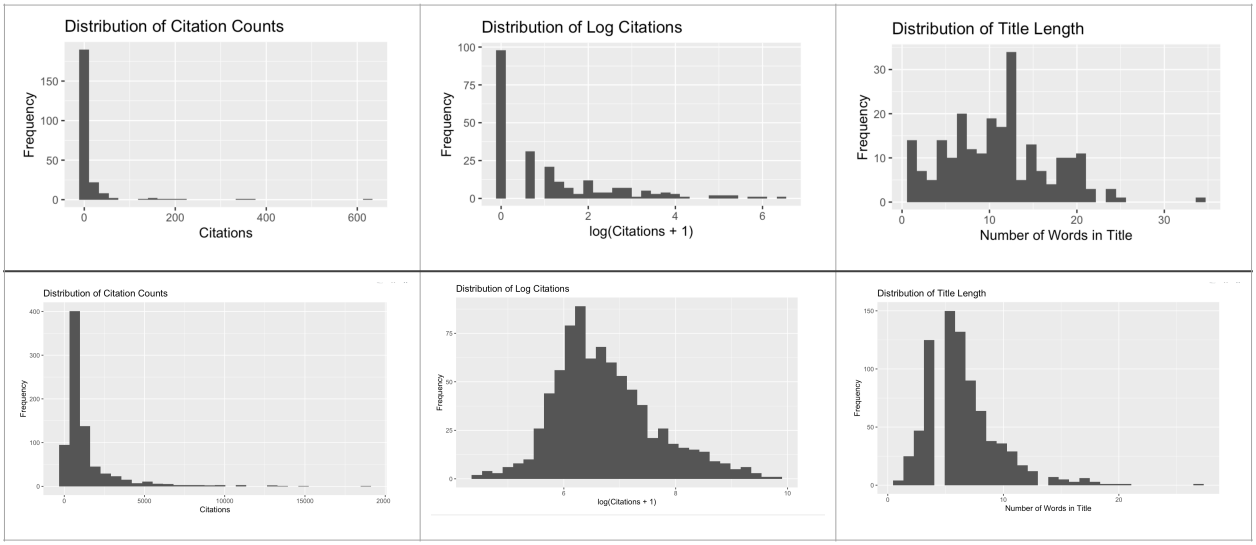
The primary dataset consists of 231 economics papers published between 2015 and 2020, collected from OpenAlex [3]. OpenAlex provides open-access metadata on academic publications, including citation counts and publication characteristics, but the papers themselves vary in whether their full texts are freely available. For each paper, the dataset includes the total number of citations, publication year, number of authors, open-access status, and detailed title information.

To address concerns about sample size and data-source dependence, I construct a second dataset using economics papers obtained from Semantic Scholar [4]. This expanded dataset contains nearly 800 papers and represents the first set of economics-related results returned by the Semantic Scholar search engine. Because all papers in the Semantic Scholar dataset are accessible through the platform, open-access status does not vary and is therefore excluded from the regression analysis for this sample.

Across both datasets, citation counts are highly right-skewed. To reduce the influence of extreme values and make the data more suitable for linear regression, the dependent variable is defined as the natural logarithm of citations plus one, log(citations + 1). Title complexity is measured using several variables constructed from the paper titles: title length, defined as the number of words in the title; average word length, defined as the average number of letters per

word; and indicator variables for whether the title contains a colon or a question mark. These measures capture different aspects of how detailed or stylistically complex a title may appear.

Distributions for OpenAlex & Semantic Scholar



Histogram of Citation Counts, Log Citations, and Title Length

3. Methodology

To examine the relationship between title complexity and citation outcomes, I estimate a sequence of Ordinary Least Squares (OLS) regression models. The dependent variable is the natural logarithm of citations plus one, log(citations + 1), which accounts for the highly right-skewed distribution of citation counts.

3.1 Baseline Specification

The baseline model relates citation outcomes to title length alone:

$$\log(\text{citations}_i + 1) = \beta_0 + \beta_1 \text{TitleLength}_i + u_i$$

where (TitleLength_i) is the number of words in the paper's title. The coefficient (β_1) measures the percentage change in citations associated with a one-word increase in title length.

3.2 Extended Model with Title Features and Controls

I then estimate a more complete specification that includes additional title characteristics and control variables:

$$\log(\text{citations}_i + 1) = \beta_0 + \beta_1 \text{TitleLength}_i + \beta_2 \text{AvgWordLength}_i + \beta_3 \text{Colon}_i + \beta_4 \text{Question}_i + \beta_5 \text{NumAuthors}_i + \beta_6 \text{Year}_i + u_i$$

In the OpenAlex dataset, I additionally include an indicator for open-access availability. In the Semantic Scholar dataset, this variable is excluded because it does not vary across observations.

3.3 Nonlinear Specification

To test whether the effect of title length is nonlinear, I estimate the following model:

$$\log(\text{citations}_i + 1) = \beta_0 + \beta_1 \text{TitleLength}_i + \beta_2 \text{TitleLength}_i^2 + X_i'\gamma + u_i$$

where (X_i) includes the same set of control variables. This specification allows the marginal effect of title length to vary with title length.

3.4 Hypothesis Testing and Inference

For each coefficient, I conduct two-sided t-tests with the null hypothesis $(H_0 : \beta_j = 0)$. Overall model significance is assessed using F-tests, and model fit is evaluated using R-squared values. All models are estimated separately for each dataset to allow for comparison and replication of results.

Regression Models for OpenAlex & Semantic Scholar

```
Call:
lm(formula = log_citations ~ title_length, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-1.3320 -1.1482 -0.4550  0.7893  5.1710

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.34868    0.19965   6.755 1.16e-10 ***
title_length -0.01671    0.01605  -1.041    0.299
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.43 on 229 degrees of freedom
Multiple R-squared:  0.004709,  Adjusted R-squared:  0.0003626
F-statistic: 1.083 on 1 and 229 DF,  p-value: 0.299
```

```
Call:
lm(formula = log_citations ~ title_length + avg_word_length +
    has_colon + num_authors + publication_year + open_access_dummy,
    data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-2.7294 -0.9809 -0.3032  0.6170  4.3336

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      266.36034   35.05648   7.598 8.13e-13 ***
title_length      -0.01674    0.01753  -0.955  0.34049
avg_word_length   -0.03186    0.07037  -0.453  0.65114
has_colon          0.07617    0.18805   0.405  0.68581
num_authors        0.03160    0.02067   1.529  0.12764
publication_year  -0.13126    0.01737  -7.555 1.06e-12 ***
open_access_dummy  0.52365    0.17275   3.031  0.00272 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.263 on 224 degrees of freedom
Multiple R-squared:  0.2407,    Adjusted R-squared:  0.2204
F-statistic: 11.83 on 6 and 224 DF,  p-value: 1.596e-11
```

```
Call:
lm(formula = log_citations ~ title_length + I(title_length^2) +
    avg_word_length + has_colon + num_authors + publication_year +
    open_access_dummy, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-2.7766 -0.9150 -0.3430  0.6492  4.2991

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      266.410423  35.072426   7.596 8.33e-13 ***
title_length       0.025740   0.050717   0.508  0.61229
I(title_length^2)  -0.001625   0.001821  -0.893  0.37298
avg_word_length    -0.015923   0.072632  -0.219  0.82667
has_colon          0.047849   0.190793   0.251  0.80221
num_authors        0.028945   0.020891   1.386  0.16726
publication_year  -0.131425   0.017383  -7.561 1.04e-12 ***
open_access_dummy  0.502698   0.174413   2.882  0.00433 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.264 on 223 degrees of freedom
Multiple R-squared:  0.2434,    Adjusted R-squared:  0.2196
F-statistic: 10.25 on 7 and 223 DF,  p-value: 4.012e-11
```

```
Call:
lm(formula = log_citations ~ title_length, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1934 -0.5950 -0.1458  0.4736  3.1154

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.64082    0.07458  89.039   <2e-16 ***
title_length  0.01496    0.01026   1.457    0.145
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9097 on 795 degrees of freedom
Multiple R-squared:  0.002665,  Adjusted R-squared:  0.00141
F-statistic: 2.124 on 1 and 795 DF,  p-value: 0.1454
```

```
Call:
lm(formula = log_citations ~ title_length + avg_word_length +
    has_colon + has_question + num_authors + publication_year,
    data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1297 -0.6125 -0.1422  0.5020  3.1542

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      34.004493   3.644176   9.331  < 2e-16 ***
title_length      0.020052   0.012450   1.611 0.107668
avg_word_length  -0.008270   0.026576  -0.311 0.755736
has_colon         0.082401   0.081276   1.014 0.310967
has_question     -0.692188   0.191108  -3.622 0.000311 ***
num_authors       0.005999   0.010973   0.547 0.584722
publication_year -0.013702   0.001832  -7.479 1.99e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8722 on 790 degrees of freedom
Multiple R-squared:  0.08898,   Adjusted R-squared:  0.08206
F-statistic: 12.86 on 6 and 790 DF,  p-value: 6.771e-14
```

```
Call:
lm(formula = log_citations ~ title_length + I(title_length^2) +
    avg_word_length + has_colon + has_question + num_authors +
    publication_year, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1101 -0.6136 -0.1516  0.4982  3.2709

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       34.0236598  3.6436705   9.338  < 2e-16 ***
title_length       0.0575851  0.0359691   1.601 0.109786
I(title_length^2) -0.0019862  0.0017858  -1.112 0.266387
avg_word_length   -0.0005398  0.0274660  -0.020 0.984326
has_colon          0.0651276  0.0827343   0.787 0.431407
has_question      -0.7072176  0.1915562  -3.692 0.000238 ***
num_authors        0.0092753  0.0113600   0.816 0.414469
publication_year  -0.0138075  0.0018341  -7.528 1.41e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8721 on 789 degrees of freedom
Multiple R-squared:  0.0904,    Adjusted R-squared:  0.08233
F-statistic:  11.2 on 7 and 789 DF,  p-value: 1.438e-13
```

Simple OLS Regression, Multiple Regression, and Nonlinear Regression

4. Regression Results

This section presents the regression results examining whether title complexity affects citation outcomes. All models use log(citations + 1) as the dependent variable.

4.1 Baseline Results (OpenAlex Dataset)

I begin with a simple baseline model that relates citation outcomes to title length:

$$\log(\text{citations}_i + 1) = \beta_0 + \beta_1 \text{TitleLength}_i + u_i$$

The estimated coefficient on title length is ($\hat{\beta}_1 = -0.0167$) with a p-value of 0.299. This implies that adding one word to a title is associated with approximately a 1.7% decrease in citations on average, but the effect is not statistically significant. The model explains less than 1% of the variation in citation counts ($R^2 = 0.0047$), indicating that title length alone has very little explanatory power.

I then estimate a full multiple regression that includes additional title characteristics and control variables. In this specification, title length, average word length, and colon usage all remain statistically insignificant. In contrast, publication year has a large and statistically significant negative coefficient (($\hat{\beta} \approx -0.13$), $p < 0.001$), reflecting that newer papers have had less time to accumulate citations. Open-access status is positive and statistically significant (($\hat{\beta} \approx 0.52$), $p < 0.01$), suggesting that open-access papers receive substantially more citations. The full model explains about 24% of the variation in citations ($R^2 \approx 0.24$).

To test for nonlinear effects, I include a squared term for title length. Neither the linear nor the quadratic term is statistically significant, and the R^2 changes very little. This provides no evidence of a nonlinear relationship between title length and citations in the baseline dataset.

4.2 Replication Results (Semantic Scholar Dataset)

I replicate the same analysis using the larger Semantic Scholar dataset. Consistent with the baseline results, title length, average word length, and colon usage remain statistically insignificant across all specifications. Their estimated coefficients are small in magnitude, and R^2 values remain modest (approximately 0.09 in the full model).

One notable difference emerges for question-style titles. In the full model, the coefficient on the question mark indicator is negative and statistically significant (($\hat{\beta} \approx -0.69$), $p < 0.001$). This implies that papers with question-style titles receive substantially fewer citations on average, even after controlling for authorship and publication year. This result is also visible in the descriptive boxplots (Appendix Figures A4) and remains robust when allowing for nonlinear effects.

Publication year continues to be the strongest predictor of citations in the replication sample (($\hat{\beta} \approx -0.014$), $p < 0.001$), confirming that citation counts are strongly driven by exposure time rather than title characteristics.

5. Robustness and Diagnostics

To assess the robustness of the main findings, I conduct several additional checks and examine standard regression diagnostics. These analyses evaluate whether the results are sensitive to model specification or violate key OLS assumptions.

5.1 Robustness Checks

First, I test whether the relationship between title length and citations is nonlinear by adding a squared term for title length to the regression model:

$$\log(\text{citations}_i + 1) = \beta_0 + \beta_1 \text{TitleLength}_i + \beta_2 \text{TitleLength}_i^2 + X_i'\gamma + u_i$$

Across both datasets, neither the linear nor the quadratic term is statistically significant. In the OpenAlex sample, the squared term has a p-value greater than 0.30, and model fit changes minimally (R^2 increases from approximately 0.24 to 0.243). Similarly, in the Semantic Scholar dataset, the inclusion of the quadratic term does not meaningfully improve explanatory power ($R^2 \approx 0.09$). These results indicate no evidence of a nonlinear relationship between title length and citation outcomes.

Second, I compare results across datasets collected from different sources. The insignificance of title length, average word length, and colon usage is consistent in both the OpenAlex and Semantic Scholar samples, suggesting that the null results are not driven by a particular dataset or sample size. The negative and statistically significant effect of question-style titles appears

only in the larger Semantic Scholar dataset, indicating that this relationship becomes detectable with increased statistical power.

5.2 Diagnostic Checks

Standard diagnostic plots are examined to assess the validity of the OLS assumptions. Residuals-versus-fitted-value plots show no systematic patterns, supporting the linearity assumption. Residuals plotted against title length also display no visible structure, consistent with the estimated insignificance of title length.

Normal Q–Q plots indicate that residuals are approximately normally distributed, with mild deviations in the tails. Given the sample sizes and the log transformation of citations, these deviations are unlikely to substantially affect inference. Overall, the diagnostic results suggest that the linear regression models are appropriate for the data and that the main findings are not driven by model misspecification.

(Diagnostic plots are reported in Appendix Figures A1–A3.)

6. Econometric Concerns and Limitations

While the empirical analysis provides useful evidence on the relationship between title complexity and citation outcomes, several econometric concerns and limitations should be acknowledged.

First, omitted variable bias may affect the estimated coefficients. Citation counts are influenced by many factors that are not observed in the data, such as journal quality, author reputation, institutional affiliation, research topic, and paper quality. If these omitted factors are correlated with title characteristics, the estimated effects may be biased. Although controlling for publication year and number of authors helps address some variation, it is not possible to fully account for all determinants of citations in this setting.

Second, the analysis does not establish causality. The regression results describe correlations between title features and citation outcomes, but they do not imply that changing a paper's title would directly cause citations to increase or decrease [2]. Authors may choose certain title styles based on the nature or importance of the research, which could confound interpretation. For example, papers with question-style titles may differ systematically in content or contribution, rather than the title format itself driving citation outcomes.

Third, measurement limitations may affect the construction of key variables. Title complexity measures such as title length and average word length are simple proxies and may not fully capture how readable or informative a title appears to readers [2]. Similarly, citation counts reflect both academic impact and exposure time, and even with publication year controls, differences in citation practices across subfields may remain.

Finally, sample selection may limit generalizability. Both datasets focus on economics papers indexed by OpenAlex and Semantic Scholar, which may not be representative of all academic disciplines or publication outlets. As a result, the findings may not generalize beyond economics or to papers not captured by these databases.

Despite these limitations, the consistency of results across two independently collected datasets provides confidence that the main conclusions are not driven by a particular sample or specification.

7. Conclusion

This paper examines whether the stylistic complexity of academic paper titles influences citation outcomes in economics. Using two independently collected datasets and a series of regression models, I analyze several common measures of title complexity, including title length, average word length, and the use of punctuation such as colons and question marks. Citation counts are modeled using a log transformation to account for skewness, and results are compared across specifications and data sources.

Across both datasets, the findings consistently show that most measures of title complexity have little to no meaningful relationship with citation counts. Title length, word complexity, and colon usage are not statistically significant predictors of citations once basic controls are included. These results suggest that commonly debated stylistic choices in titles do not play a major role in determining academic impact. However, one notable exception emerges in the larger Semantic Scholar dataset: papers with question-style titles receive significantly fewer citations on average. This finding suggests that certain framing choices may be associated with reduced visibility or impact, although the analysis does not establish causality.

Overall, the results indicate that citation outcomes are driven more by structural factors, such as publication timing, than by title complexity. While authors may devote considerable effort to choosing a title, the evidence from this study suggests that most stylistic features have limited influence on citation performance. Future research could explore whether these patterns differ across subfields, journals, or disciplines, or examine alternative measures of title readability and framing.

Appendix A: Additional Figures

Figure A1: Residuals vs fitted values (OpenAlex & Semantic Scholar)

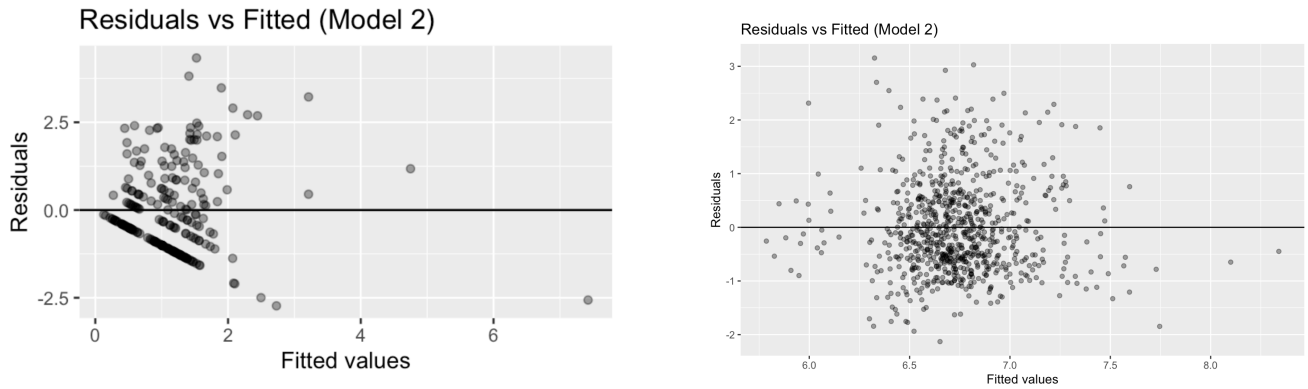


Figure A2: Normal Q–Q plot of residuals (OpenAlex & Semantic Scholar)

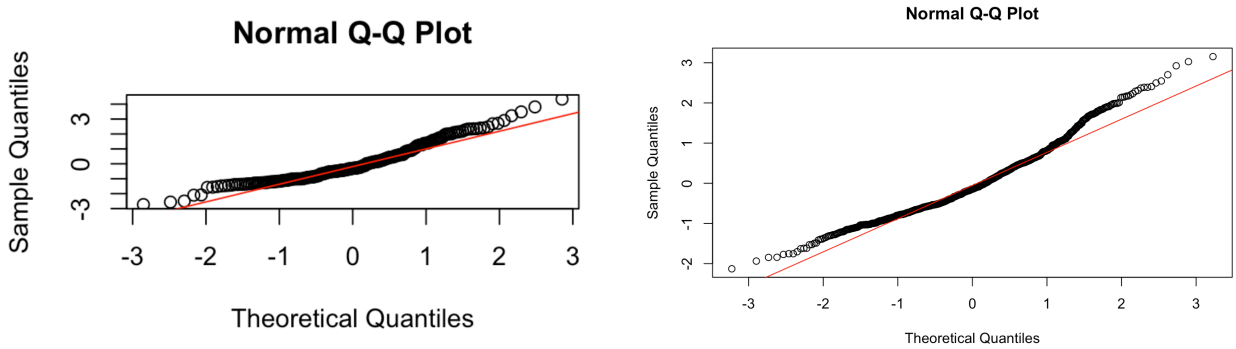


Figure A3: Residuals vs title length (OpenAlex & Semantic Scholar)

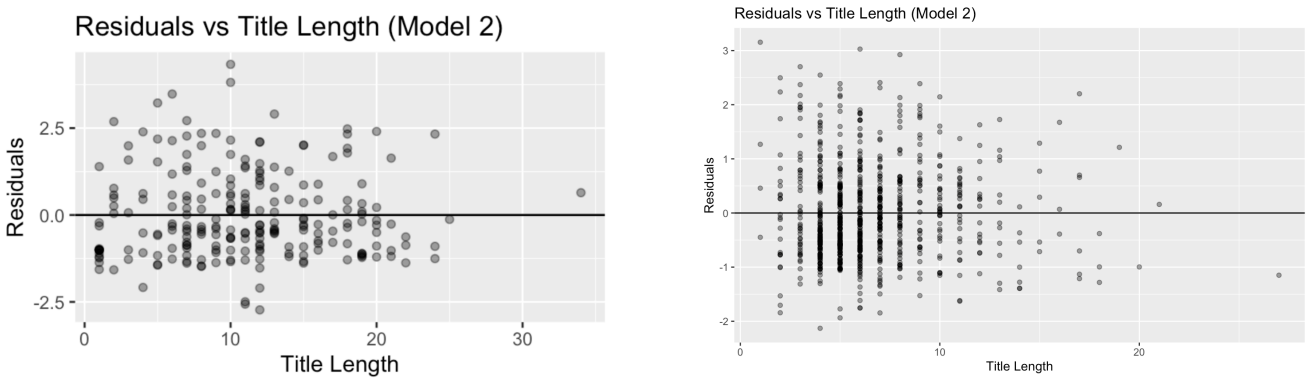
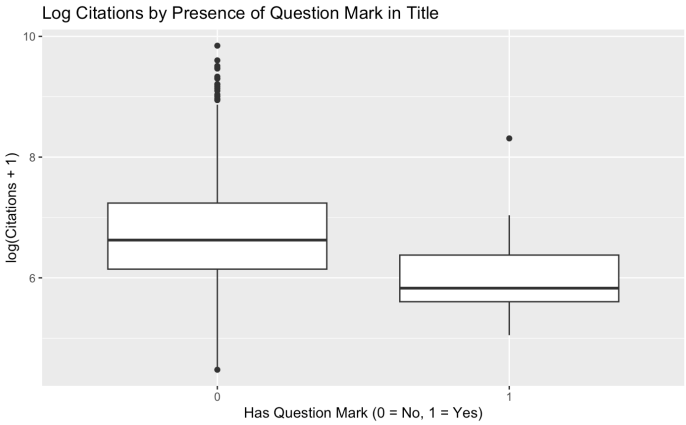


Figure A4: Boxplot of Question Mark (Semantic Scholar)



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